

Arrhythmia

Learning Objectives

By the end of this lecture, students will be able to:

Define what an arrhythmia is.

Describe normal cardiac conduction and rhythm.

Identify common types of arrhythmias.

Recognize signs, symptoms, and risk factors.

Discuss nursing assessment and management for patients with arrhythmias.

Definition

An arrhythmia is an abnormality in the rate, rhythm, or sequence of cardiac electrical impulses.

Normal heart rate: 60–100 beats per minute (bpm)

Normal rhythm: Sinus rhythm — impulse originates from the sinoatrial (SA) node

Normal Cardiac Conduction System

Pathway:

SA node → AV node → Bundle of His → Right & Left Bundle Branches → Purkinje fibers

Key concept:

Electrical activity triggers **mechanical contraction** of the heart. Disturbance in this system = arrhythmia.

Classification of Arrhythmias

Type	Example	Heart Rate	Description
Bradyarrhythmias	Sinus bradycardia, AV block	<60 bpm	Slow rhythm
Tachyarrhythmias	Sinus tachycardia, SVT, VT	>100 bpm	Fast rhythm
Irregular rhythms	Atrial fibrillation, PVCs	Variable	Irregular beat pattern

Common Arrhythmias and ECG Features

A. Sinus Bradycardia

- HR < 60 bpm, regular rhythm
- Often normal in athletes or during sleep
- **Causes:** Vagal stimulation, medications (beta-blockers, digoxin)
- **Treatment:** If symptomatic → Atropine, oxygen, IV fluids

B. Sinus Tachycardia

- HR > 100 bpm, regular rhythm
- **Causes:** Fever, pain, anxiety, dehydration, heart failure
- **Treatment:** Treat the underlying cause

C. Atrial Fibrillation (AFib)

- Irregularly irregular rhythm, no P waves
- **Risk:** Stroke, thromboembolism
- **Symptoms:** Palpitations, fatigue, shortness of breath
- **Management:**
 - Rate control: Beta-blockers, calcium channel blockers
 - Anticoagulation: Warfarin, DOACs
 - Cardioversion if indicated

D. Atrial Flutter

- “Sawtooth” pattern on ECG
- Often regular, 250–350 bpm atrial rate
- **Management:** Similar to AFib

E. Ventricular Tachycardia (VT)

- Wide QRS complexes, rate >100 bpm
- **Life-threatening!**
- **Causes:** Myocardial infarction, electrolyte imbalance
- **Treatment:**
 - With pulse: Amiodarone, synchronized cardioversion
 - Pulseless: CPR + Defibrillation

F. Ventricular Fibrillation (VF)

- Chaotic, no organized rhythm or cardiac output
- **Emergency!**
- **Management:** CPR, Defibrillation immediately

G. Asystole

- Flat line (no electrical activity)
- **Management:** CPR, Epinephrine, treat reversible causes (H's & T's)

5. Nursing Assessment

- Monitor vital signs and ECG rhythm strips
- Assess for signs of decreased cardiac output:
 - Hypotension
 - Dizziness
 - Chest pain
 - Shortness of breath
 - Altered mental status
- Check medications, electrolytes, and oxygen levels

6. Nursing Management & Interventions

1. Maintain airway, breathing, circulation (ABCs)
2. Apply oxygen as ordered
3. Continuous ECG monitoring
4. Administer antiarrhythmic drugs as prescribed
5. Prepare for emergency procedures:
 1. Defibrillation
 2. Cardioversion
 3. Temporary pacing

7. Complications

Decreased cardiac output

Thromboembolism (especially in AFib)

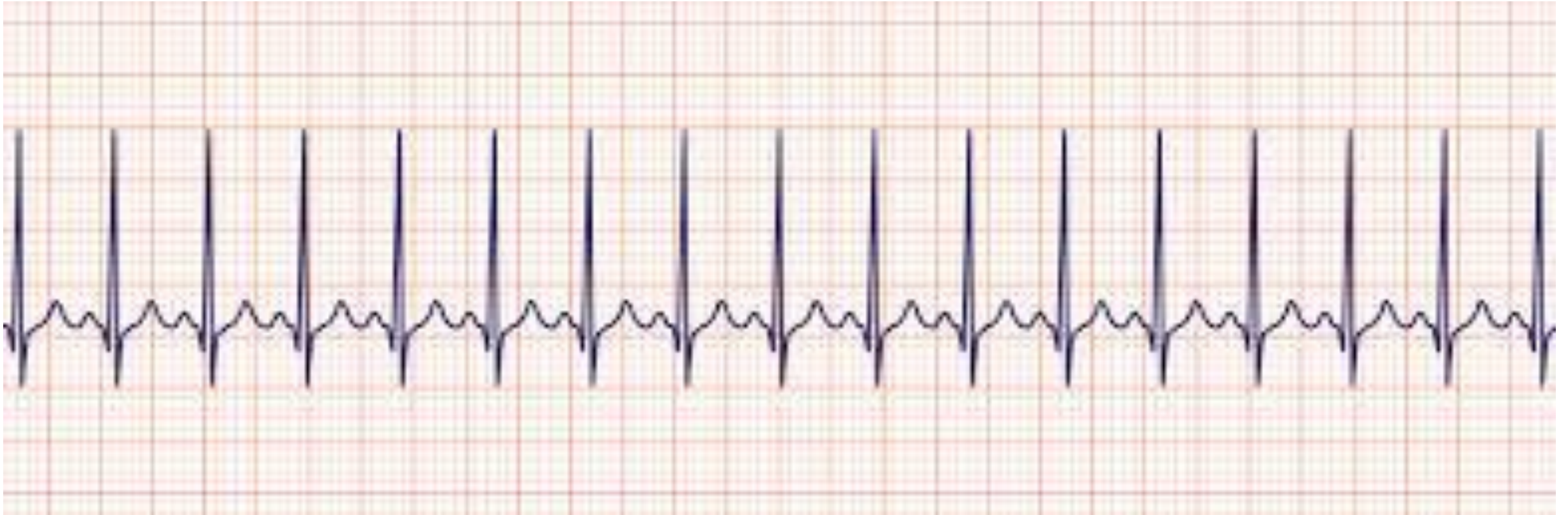
Sudden cardiac arrest

Heart failure

8. Summary

- Arrhythmias are disturbances in heart rhythm.
- ECG interpretation is key to diagnosis.
- Prompt recognition and intervention save lives.
- Nurses play a crucial role in monitoring, treatment, and patient education.

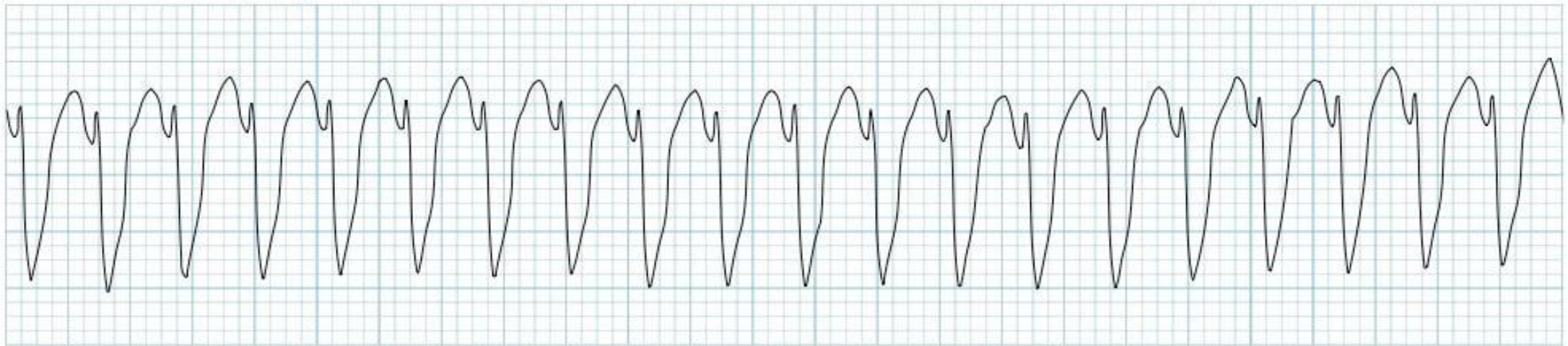
Sinus tachycardia



Atrial fibrillation



Ventricular tachycardia



Thank You